

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Design) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Computer Science and Design) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTCOC304 - COMPUTER ARCHITECTURE & ORGANIZATION

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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1. Computer address bus is -
a. Multidirectional b. Bidirectional c. Unidirectional d. No Direction
2. Which of the architecture is power efficient?
a. RISC b. ISA c. IANA d. CISC
3. Which of the following computer registers collects the result of computation?
a. Accumulator b. Instruction Pointer c. Storage register d. None of the above
4. The input hexadecimal representation of 1110 is _____
a. 0111 b. E c. 15 d. 14
5. To reduce the memory access time we generally make use of _____
a. SDRAM b. Heaps c. Cache d. Higher capacity RAM
6. The DMA transfers are performed by a control circuit called as _____
a. Device interface b. DMA controller c. Data controller d. Overlooker
7. The method of accessing the I/O devices by repeatedly checking the status flags is _____
a. Program-controlled I/O b. Memory-mapped I/O
c. I/O mapped I/O d. None of the mentioned
8. The CPU Consists of _____
a. ALU and Control unit only
b. ALU , Control unit and Registers only
c. ALU , Control unit and system bus only
d. ALU , Control unit, Registers and Internal Bus
9. What is the high speed memory between the main memory and the CPU called?
a. Register Memory b. Cache Memory c. Storage Memory d. Virtual Memory
10. The method generally used in computers for subtraction is?
a. 1's complement b. 2's complement c. 9's complement d. None of the above
11. Which part of the computer is used for calculating and comparing ?
a. ALU b. Control unit c. Disk unit d. Modem
12. The collection of 8-bits is called as -
a. Byte b. Nibble c. Word d. Record

Q2. Solve the following.

- A) Explain the basic functional unit of the computer and explain each unit in detail.

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- B) Discuss about Bus structure with a neat sketch? 6
- Q3. Solve the following.
- A) Explain the design of ALU in detail 6
 - B) Difference between RISC and CISC 6
- Q4. Solve Any Two of the following.
- A) Subtract $(11011)_2 - (10011)_2$ using 1's complement and 2's complement method 6
 - B) Write the 2's complement of 1011011 6
 - C) Convert given binary number $(1010.01)_2$ to decimal number 6
- Q5. Solve Any Two of the following.
- A) Explain cache memory in detail? 6
 - B) Compare SRAM and DRAM. 6
 - C) Explain Memory Hierarchy level in detail 6
- Q6. Solve Any Two of the following.
- A) Differentiate Programmed I/O and Interrupt I/O 6
 - B) In virtually all systems that include DMA modules, DMA Access to main memory is given priority over CPU access to main memory. Why? 6
 - C) What is difference between hardwired implementation and a microprogrammed implementation of control unit 6

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